Project Design Phase-I

Solution Fit

Parameter	Details
Date	08 March 2026
Team ID	
Project Name	VoltEdge: EV Analytics Portal
Maximum Marks	2 Marks

Define CS, fit into CC	CUSTOMER SEGMENT(S) [CS]	CUSTOMER CONSTRAINTS [CC]	AVAILABLE SOLUTIONS [AS]	Explore AS, differentiate
	• **Urban Energy Planners (e.g., Aarav):** Needs to manage grid load and plan public charging infrastructure. • **EV Fleet Managers (e.g., Meera):** Needs to maintain vehicle uptime and monitor battery health for a large taxi/delivery fleet. • **Prospective EV Buyers (e.g., Ravi):** Needs unbiased data to compare models, range, and long-term savings.	• **Data Fragmentation:** Relevant EV data is scattered across different manufacturer websites. • **Technical Complexity:** Real-time battery telemetry is hard for non-technical managers to interpret. • **Budget:** Limited funds for city planning require high-accuracy data before building	• Manual logs and spreadsheets (Time-consuming and prone to human error). • Manufacturer brochures (Often biased or "ideal condition" data only). • Generic GPS tracking (Doesn't provide electrical/battery health depth).	

		stations.		
Focus on J&P, tap into BE, understand RC	**JOBS-TO-BE-DONE / PROBLEMS [J&P]** • Identifying "hotspots" where charging demand is highest to avoid grid failure. • Tracking **State of Charge (SoC)** and **Depth of Discharge (DoD)** to predict battery degradation. • Comparing different drivetrain types (AWD vs RWD) and body styles (SUV vs Sedan) for energy efficiency.	**PROBLEM ROOT CAUSE [RC]** The lack of a standardized, open-access analytics layer for the EV ecosystem. As vehicles become "computers on wheels," the volume of data exceeds what traditional manual methods can handle.	**BEHAVIOUR [BE]** • Customers currently spend hours manually cross-referencing CSV files or visiting multiple dealership websites to calculate potential savings.	**Focus on J&P, tap into BE, understand RC**
Identify strong TR & EM	**TRIGGERS [TR]** • A fleet manager seeing a sudden drop in vehicle range. • A city planner receiving reports of power outages near unmanaged charging clusters. • A consumer seeing rising fuel prices and looking for an efficient electric alternative. **EMOTIONS: BEFORE/AFTER [EM]** • **Before:** Confused, anxious about "range anxiety,"	**YOUR SOLUTION [SL]** **VoltEdge Analytics Portal:** • **Backend:** A robust **PostgreSQL** database that centralizes disparate EV datasets. • **Visualization:** Interactive **Tableau** dashboards (embedded in your index.html) that turn raw rows into heatmaps and efficiency leaderboards. • **Accessibility:** A Flask-driven web interface that allows users to filter data by	**CHANNELS OF BEHAVIOUR [CH]** • **Online:** Searching for "Best EV India" or "EV charging maps." • **Offline:** Consulting with energy grid technicians or attending automotive expos.	**Identify strong TR & EM**

	overwhelmed by technical specs. • **After:** Confident, in control of fleet health, and informed about infrastructure investments.	Brand, Price, and PowerTrain in seconds.		